The two earlier volumes used numbers as familiar objects. Their arithmetic and order were school facts, named where they were used and never proved. This volume proves them. It starts from Peano's axioms for the natural numbers and builds the integers, the rationals and the real numbers, each out of the one before. At the end the real numbers form a complete ordered field, and any two complete ordered fields are the same up to exactly one isomorphism.

Who it is for. The reader we have in mind has worked through *Logic and Proof* and *Sets and Functions*, or has their habits from elsewhere. That means negating a quantified statement, running an induction, and checking that a rule on equivalence classes does not depend on the representative chosen. The last of these is used in almost every chapter. The chapters point back to both volumes where they use a definition or a rule, so a reader who is unsure of a step can look it up.

What it covers.

Chapter	Title	What it does
1	The Natural Numbers	$\mathbb{N}$ as a Peano system: induction, the recursion theorem, addition and multiplication, order and well-ordering, any two Peano systems alike; a short closing section on where $\mathbb{N}$ comes from in set theory
2	The Integers	$\mathbb{Z}$ as classes of pairs of natural numbers, well-defined operations, commutative rings, ordered rings and the rules for inequalities, division with remainder
3	The Rationals	$\mathbb{Q}$ as classes of fractions, an Archimedean ordered field, and the gap in $\mathbb{Q}$ where $\sqrt{2}$ should be
4	Ordered Fields and Sequences	Limits and Cauchy sequences in an arbitrary ordered field, the Archimedean property, and a Cauchy sequence of rationals with no rational limit
5	The Real Numbers from Cauchy Sequences	$\mathbb{R}$ as classes of rational Cauchy sequences, an ordered field in which every bounded set has a supremum

6	The Complete Ordered Field Is Unique	Exactly one order-preserving isomorphism between any two complete ordered fields, and what *Analysis on the Real Line* takes from this volume

Chapters 1–3 build the numbers that algebra needs, one chapter for each. Chapters 4 and 5 fill the gap that Chapter 3 found in $\mathbb{Q}$: first the rules for sequences, proved without the number they aim at, then the numbers made from the sequences themselves. Chapter 6 shows that the construction does not matter.

How to read it. The main line is Chapters 1–6; read them in order, since each uses the ones before it. Every section has Quick checks with folded answers; answer each one before you unfold it. The exercises of a section come in three groups, A to check understanding, B for practice and C to go deeper, and the last section of each chapter adds R items that reach back to earlier chapters. A chapter's opening page lists what it needs, with links, and the time it takes. Second-reading folds, in most chapters, hold edge cases and longer justifications. They can stay closed on a first reading, but they are not optional, since later proofs may use what they contain. Results marked optional can be skipped, and nothing outside them depends on them.

Time. About 120 hours on the core track (all of A, one B and one C per section, the exit checks) and about 180 hours for everything. Chapter by chapter, core and full: Chapter 1, 20 and 30 hours; Chapter 2, 20 and 25; Chapter 3, 15 and 25; Chapter 4, 25 and 40; Chapter 5, 25 and 35; Chapter 6, 15 and 25. The figures come from word counts and exercise counts, not from reader logs, and will be replaced by the median of logged hours once there are some.

What comes next. *Analysis on the Real Line* does not need this volume. It starts from the axioms of a complete ordered field, the axioms this volume proves for its $\mathbb{R}$, and Chapter 6 explains why it does not matter which model of them one has in mind. A reader who wants to know that those axioms describe something real, and only one thing, will find both answers here. *Algebraic Preliminaries* builds on Chapters 2, 5 and 6: the integers with division with remainder (its Chapter 1 takes up divisibility, prime factorization and congruences), the real numbers as a complete ordered field, and square roots in it, together with the ring and field rules of Chapters 2, 3 and 4 that come with them.